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Numerical Prediction of Delamination Propagation in Laminated Composite Material under Mode-I Loading

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Declaration

This is to certify that the project and thesis work entitled *"Numerical Prediction of Delamination Propagation in Laminated Composite Material under Mode-I Loading"* has been carried out by *Agnila Ghosh Surovi, Roll No. 1627024* in the Department of Materials Science and Engineering, Khulna University of Engineering & Technology, Khulna, Bangladesh. The above thesis work or any part of this work has not been submitted anywhere for the award of any degree or diploma.

Signature of Supervisor	Signature of Student
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At first, I want to praise my God for always being with me and giving me the strength to overcome all the difficulties.

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Abstract

Composite materials are the most appealing choice for aeronautical and automobile applications nowadays because of their high strength with light weight. The most common defect in laminated composites is delamination, so it becomes necessary to analyze delamination in laminated composites to prevent structures from the defect during service. The experimental investigation of delamination is not always possible, so efficient numerical modeling techniques for delamination analysis are necessary to identify. Various modeling approaches such as continuum damage model, floating node method, virtual crack closure technique, and cohesive zone modeling have been used previously. However, more inquiry is required to find out the suitable modeling method for the study of delamination. This research has modeled a double cantilever beam using the virtual crack closure technique, the cohesive surface technique, and the cohesive element method. The effect of orientation on delamination propagation is analyzed. From the view of computational time and accuracy, the VCCT technique is the most practical for delamination analysis. Cohesive surface and cohesive element methods follow energy conservation law in calculating energy dissipation. In the case of orientation, 45-degree ply orientation is the optimum for both better delamination prevention and better tensile properties.

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Nomenclature

Symbols and Acronyms	Description
G	Energy release rate
G_c	Critical energy release rate or fracture toughness
W	Virtual work done
F	Force close to the crack
δ_u	Displacement for the upper node
δ_l	Displacement for the lower node
ΔA	Created crack surface area
b	Width of the crack
Δa	Crack length
G_I, G_{II} and G_{III}	Energy release rate for mode-I, mode-II and mode III respectively
G_c	Cohesive fracture energy
σ_c	Cohesive strength of the material
t_n, t_s and t_t	Nominal cohesive stresses for mode-I, mode-II and mode III respectively
t_n^0, t_s^0 and t_t^0	Peak value of nominal cohesive stresses for mode-I, mode-II and mode III respectively
G_T	Sum of the strain energy release rate
η	B-K material parameter
G_I, G_{II} and G_{III}	Critical energy release rate for mode-I, mode-II and mode III respectively
δv	Kinematically admissible virtual displacement
x_i^a	Coordinates of unique points

u_i^a	Displacement vector at each nodal point
$N^a(x)$	Shape function
δv_i^a	Arbitrary nodal values of virtual velocity
K_{aibk}	Global stiffness matrix
k_{aibk}	Element stiffness matrix
ξ_i	Normalized coordinates
f_i	Force matrix
D	Damage internal variable
η_i	Kinematically admissible displacement field

CHAPTER I

Introduction

1.1 Overview

Composite is an anisotropic heterogeneous structure that consists of three phases: primary phase, secondary phase, and interphase [1, 2]. Generally, matrix constitutes the preliminary phase, fiber or any reinforcement constructs the secondary phase, and interphase ensures adhesion between matrix and fiber [2]. Fiber or reinforcement imparts strength and stiffness to the composite material. The matrix holds the fiber or reinforced phase. The adhesion between fiber and matrix ensures the uniform load transfer to the fiber from the matrix. Composites provide high strength, stiffness, corrosion resistance and design flexibility with a facility of low density; however, there is a need to develop some properties of composite like- thermal and electrical conductivity, thermal expansion, swelling, and shrinkage. Due to appreciable mechanical performance and fatigue withstand capability, the use of composite materials in aerospace and automotive applications increases day by day [3].

The manufacturing process leaves some defects in composite materials, such as resin-rich areas within an individual laminate or at a laminate interface; voids; distorted fibers; broken fibers; inclusions [2]. Besides, composite materials have numerous failure mechanisms, such as fiber-matrix debonding, fiber pull-out, intralaminar matrix cracking, matrix deformation, delamination, and fiber fracture [4–8].

The fundamental mode of damage in the laminated composite is delamination. Delamination is the detachment through the interfaces of the laminates due to weaker interlaminar strength triggered by interlaminar tensile loading, shear loading, in-plane quasi-static loading, fatigue loading, and impact loading [4, 6, 9]. Delamination arises several situations that are difficult to handle [10]. Firstly, delamination can reduce the structural integrity, compressive strength, and load-carrying capacity of composite materials. Secondly, many stimuli during the service life of composite can stimulate delamination [11]. Finally, delamination can't be inspected quickly [12]. However, delamination around a sharp notch can redistribute the stress field that isolates the defect [13]. Delamination is anticipated by damage mechanics based on maximum allowable stress or strain [14]. The simulation of delamination in the composite can be performed using the finite element method (FEM) by implying the virtual crack closure technique (VCCT) and cohesive finite elements. In this study, a double cantilever beam specimen has been modeled using the VCCT, cohesive element, and cohesive surface technique, and delamination propagation for each model has been simulated.

1.2 Motivation

Previous researches provide concepts for understanding the damage mechanism of delamination [15], complex delamination formation by impact loading [16], fatigue loading [17], matrix cracking, and delamination migration [18], suggesting damage criteria for delamination estimation [19], determining delamination toughness [20], and identifying the delamination process [21]. From an experimental perspective, the development of delamination testing methodology in the composite is visible, such as the delamination fracture toughness test [22] and the delamination analysis under impact loading [23, 24]. From a simulation point of view, the methodology for modeling delamination accurately and effectively is developing continuously. On top of that, the development of available approaches such as the virtual crack closure technique (VCCT), the cohesive zone model (CZM) [25], the floating node method (FNM) [26], and the continuum damage model (CDM) [27, 28] can be observed. However, the gap in the study of comparison and combination of these methods is not covered in the literature. A combined analysis of delamination using different modeling techniques by providing the result from each modeling approach is absent in the previous studies. The representation of these modeling procedures, advantages, disadvantages, similarities, and contrast hasn't been studied earlier. Hence, this study has established a detailed comparative view of VCCT and CZM to overcome the insufficiency of knowledge.

1.3 Objectives

This thesis research aims to develop finite element models for delamination analysis in laminated composite material as laminated composites are susceptible to delamination. The purposes of this study are given below:

- To compare the existing finite element modeling techniques and determine the efficient usage of different modeling approaches for delamination initiation and propagation analysis.
- To investigate the similarity of critical load required for delamination obtained from the simulation and the analytical solution.
- To analyze the variation of computational time of three different approaches.
- To determine the dissimilarity of dissipated energy calculation in these three approaches.
- To understand the relationship between crack length and critical displacement in three different approaches.
- To examine the effect of material orientation on delamination characteristics.
- To assess the simulation results and suggest advanced future analysis.

1.4 Organization of Thesis

The content of this thesis report is organized as follows. Chapter 2 represents a review study on the previous experimental, analytical, and numerical research on delamination. In chapter 3, the Introduction of the finite element method and definition and theory of VCCT and CZM s presented. Chapter 4 illustrates the modeling steps and methods. Chapter 5 presents the effect of mode-I loading, energy, and material orientation on delamination propagation in VCCT, cohesive surface and cohesive element models, the validation of numerical solution with analytical solution. Chapter 6 concludes the analysis and dictates further studies.

CHAPTER II

Review of Literature

Researchers have performed considerable research on delaminations' effect on the mechanical properties of reinforce-matrix composites numerically and experimentally. Talreja in 1985 [29], Lee et al. in 1989 [30], and Nguyen in 1997 [31] employed the continuum damage modeling (CDM) methodology approach for progressive damage study in laminated composites. Kutlu and Chang established one of the first analytical analyses by developing a computer code and experimental investigations to investigate composite laminates' compressive response with multiple delaminations in 1992 [32]. Meo and Thieulot presented four different modeling techniques for predicting delamination failure mode for composite using a double cantilever beam [33]. The first two approaches are based on a cohesive zone model; the third approach is based on the interface with solid elements; the fourth is based on tiebreak contact in 2005 [33].

In some researches, researchers have investigated the effect of buckling behavior of laminated composites. Kyoung et al. conducted non-linear finite element analysis to analyze buckling and post-buckling behavior of composite cross-ply laminates with multiple delaminations of different sizes and locations under compression in 1998 [34]. Chen and Sun performed finite element analysis (FEA) of compressive strength in delaminated laminates using Von Karman’s nonlinearity assumption and first-order shear deformation plate theory amalgamated with a stiffness degradation method in 1999 [35]. Hwang and Liu used the finite element method for non-linear buckling analysis of composite laminates with multiple delaminations under uniaxial compressive loading in 2001 [36]. They indicated that underneath delamination has not significantly affected the buckling stress since near-surface delaminations were more prominent in size than underneath delaminations. Short et al. analyzed the effect of the delamination geometry and position through-thickness on the compressive behavior of the glass-fiber-reinforced composites without complex effects of orthotropic material properties in 2001 as buckling is one of the crucial causes of compressive strength reduction and buckling hugely dependent on the geometry of the delamination [37]. Naik and Ramasimha developed an analytical model for compression strength determination depending on the influence of delamination location and dimension on damage tolerance of laminated plain weave fabric composites in 2001 [38]. They pointed out that damage tolerance reduces with ameliorated delamination size and decreases with increasing depth of the position of delamination initially and then increases. They compared the damage tolerance limit of plain weave carbon/epoxy composites with plain weave E-glass / epoxy composites and proposed that plain weave carbon/epoxy composites are more damage tolerant. Short et al. investigated the compressive strength of low velocity impacted flat and curved glass fiber reinforced polymer (GFRP) composites in 2002 [39]. Wang et al. studied the comparison between experimental analysis and finite element analysis of the reduction of compressive strength of glass-fiber-reinforced composite laminates with multiple delaminations compared to single delaminations in 2005 [40]. Hwang and Huang employed the finite element method for studying non-linear buckling and post-buckling characteristics of unidirectional composite with two delaminations under uniaxial compression in 2005 [41]. They encountered that if the long delamination is present near the outer surface and the short delamination is present in the inner portion, it does not affect buckling stress. In contrast, whenever the short delamination is present near the outer surface, it significantly affects buckling stress. Degeorges focused on the damage analysis of aerospace composite laminates with single and multiple delaminations, with debonding and under low-velocity impact by experimenting and finite element analysis in 2006 [42]. Suemasu et al. numerically investigated the damage mechanism of composite laminates with multiply circular delaminations under compressive loading using finite element method (FEM), and they also simulated delamination crack propagation using cohesive element in 2008 [43]. Rodman et al. presented an analytical solution for analyzing the effect of delamination number, length, position, and axial and shear strains on buckling load of composite laminates with multiple delaminations using Reissner’s beam theory in 2008 [44]. Reis et al. studied the effect of delamination on the static and fatigue load-bearing capability of carbon/epoxy composite laminates that determine the composite's strength and stiffness in 2009 [45]. They concluded that there is a slight effect of delamination's size on static strength under tensile loading, and there is no significant influence on fatigue strength under cyclic tensile loading. Aslan and Sahin determined buckling load and compressive strength of E-glass/epoxy composite laminates by applying low-velocity impact experimentally and numerically using ANSYS 11.0 in 2009. They also determined the effect of fiber orientation numerically

[46]. Ovesy et al. proposed an analytical method based on first-order shear deformation theory for analyzing buckling manners of composite laminate embedded with circular and rectangular delaminations, developed a three-dimensional (3D) finite element model for buckling analysis using ANSYS5.4 general-purpose commercial software and compared the results from the analytical model with finite element model in 2010 [47]. Craven et al. generated a finite element model of carbon fiber composite laminates with circular, elliptical, and peanut-shaped delaminations, including fiber fracture crack under compressive loading to determine buckling and post-buckling behavior and stiffness and to show the effect of delamination size and shape in 2010 [48]. Amaro et al. performed experimental analysis by doing three-point bending tests to analyze the bending behavior and numerical analysis by developing a two-dimensional cohesive mixed-mode damage model using ABAQUS software to simulate delamination propagation of carbon-fiber-reinforced composite laminates with delaminations at different locations in 2011 [49]. They found that delaminations considerably affected the bending characteristics. Liu et al. accomplished two-dimensional and three-dimensional finite element analysis to study the effect of delaminations of composites plates with two through the width delaminations under compressive loading on post-buckling properties using virtual crack closure technique (VCCT) in 2011 [20]. Mohanty et al. created a finite element model for the numerical analysis of buckling load on two-dimensional (2D) delaminated composite laminates and also explored the effect of delamination area, fiber orientations, number of layers, aspect ratios on buckling behavior of woven roving glass/epoxy composite laminates with single or multiple laminates in 2013 [50]. Liu and Zheng used the three-dimensional finite element method to analyze buckling, post-buckling, and through the width multiple delaminations of symmetric and unsymmetric, unidirectional, and angle-ply composite laminates under compression in 2013 [51]. They also operated the virtual crack closure technique (VCCT) for analyzing crack propagation. Ovesy proposed a novel layerwise theory based on first-order shear deformation theory employing Rayleigh-Ritz approximation technique to scrutinize buckling and post-buckling characteristics of composite laminates with multiple delaminations, used a finite element model using ANSYS5.4, and compared the numerical result with the analytical solution in 2015 [52]. Kharghani and Soares conceived an analytical model based on layerwise higher-order shear deformation theory to dissect the effects of the area and thickness of delamination on the deviation of a composite laminate under buckling load in 2016

[53]. Aslan and Daricik analyzed the impact of different sized multiple delaminations on tensile strength, compressive strength, flexural behavior, and buckling loads of E-glass/epoxy composites in 2016 [54]. They found that numerous delaminations with large sizes significantly affect buckling load, compressive and flexural strength; however, tensile strength does not have a notable impact due to multiple delaminations. Asaee et al. examined the buckling behavior of delaminated fiber metal laminates numerically using cohesive zone modeling in ABAQUS and experimentally in 2017 [55]. Köllner and Völlmecke established classical lamination theory (CLT) to investigate the buckling characteristics of composite laminates in 2018 [56]. Imran reviewed research on buckling analysis of composite laminates that is necessary for avoiding damage under compressive loading in 2018 [57]. Mekonnen et al. analyzed the effect of delamination size and location on buckling characteristics of composite laminates using the CZM approach in 2020 [58].

Many pieces of research on composite delamination under different loading conditions have been performed from the start of composite research. Zhu analyzed mode-I, mode-II, and mixed-mode interlaminar fracture behavior of carbon/epoxy composite laminates under quasi-static and fatigue loading using double cantilever beam (DCB) specimen, the end-notched flexure (ENF) specimen, and the single-leg bending (SLB) specimen using the virtual crack closure technique (VCCT) for ABAQUS in 2008 [59]. Turon et al. suggested a finite element analysis method of delamination propagation accurately under mixed-mode loading with cohesive elements by considering the relationship between interlaminar strengths and the penalty stiffness that leads to proper energy dissipation during delamination propagation in 2010 [21]. Sebaey et al. prevented crack jumping during delamination growth of multidirectional carbon fiber reinforced composite laminates for mode-I loading by increasing bending stiffness by defining stacking sequence in 2011 [60]. They have defined six stacking sequences numerically and validated the sequences experimentally in 2012 [61]. Sosa et al. simulated delamination propagation in a double cantilever beam of glass fiber reinforced aluminum laminate (GLARE) laminate under mode-I loading using the extended finite element method (XFEM) in 2012 [62]. Huzni et al. implemented cohesive elements to model delamination propagation in composite laminates in 2013 [63]. Liu and Islam proposed a numerical anisotropic exponential non-linear cohesive zone model of composite laminate under mixed-mode loading based on continuum damage mechanics in 2013 [64]. Zhao et al. provided an accurate numerical model for delamination growth in multidirectional composite laminates with different interfaces (0/0, 45/45, and 90/90) under mode-I and mixed-mode loadings using the cohesive element in 2014 [65]. They expressed a variable fracture toughness model as a uniform function of the crack growth length for multidirectional composite laminates to exhibit fiber bridging in multidirectional composite laminates using subroutines implanted into finite element software. Ahn and Woo researched the model's efficiency with p-convergent partial discrete-layer elements for delamination analysis of composite laminates using the virtual crack closure technique (VCCT) using isotropic plates and orthotropic plates of the double cantilever beam test specimen and the orthotropic laminated square plate specimen in 2015 [66]. Giuliese et al. performed a numerical investigation of the effect of nano-modified electrospun nanofibre on the delamination characteristics of composite laminates using the cohesive zone method in 2015 [67]. LeBlanc and LaPlante examined the effect of moistness on delamination growth in carbon fiber reinforced composite under mixed-mode loading numerically using cohesive zone-based finite element modeling and experimentally under quasi-static and fatigue loading in 2016 [68]. Pappas and Botsis analyzed intralaminar mode-I fracture in double cantilever beam of carbon fiber-reinforced composite with different thicknesses (6, 10, and 14 mm) numerically using bridging traction profile implied cohesive element model in 2016 [69]. Arrese et al. presented a method based on the relation between energy release rate and crack-tip opening displacement for delamination propagation under mode-I cohesive law for unidirectional laminated composites in 2017 [70]. Zhao et al. offered an interface-dependent analytical model of plateau fracture toughness of multidirectional carbon fiber reinforced polymer composite laminates for studying the effect of interface angle on intra-ply damage and fiber bridging during delamination propagation under mode-I loading and validated the model numerically and experimentally in 2017 [71]. In another study, Zhao et al. presented a numerical model for zigzag mode-I delamination propagation in the interfaces adjacent to a 90˚ ply of laminated composites for using extended finite element method (XFEM) in 2017 [72]. Reinoso et al. used a non-linear finite thickness cohesive interface element modeling for delamination analysis of the fiber-reinforced composite under mixed-mode loading conditions in 2017 [73]. Dongen et al. utilized a blended methodology combined with continuum damage mechanics for progressive damage analysis of composite where matrix cracks are modeled with extended finite element method (XFEM), and delaminations were presented through cohesive zone model (CZM) in ABAQUS in 2018 [74]. Ghadirdokht and Rarani investigated the effect of curvature double cantilever beam of unidirectional composite laminate on steady-state toughness and fiber bridging length numerically and experimentally under mode-I loading and simulated delamination propagation in curved double cantilever beam using virtual crack closure technique (VCCT) and cohesive zone model (CZM) in ABAQUS in 2019 [75]. Rarani and Sayedain performed a comparative study of delamination propagation for two-dimensional and three-dimensional composite double cantilever beam using Virtual crack closure technique (VCCT), cohesive zone modeling (CZM), and extended finite element method (XFEM) in 2019 [76]. Gong et al. utilized three-linear cohesive zone modeling (CZM) for analyzing the effect of fiber bridging on delamination propagation of multi-directional composite laminates using a user-subroutine UMAT in the commercial finite element software in 2020 [77]. Oliveira and Donadon proposed cohesive zone modeling for investigating delamination propagation in double cantilever beam of the composite under high-cycle fatigue loading for mode-I loading, mode-II loading, and mixed-mode loading in 2020 [78]. Karmakov et al. compared two numerical methods: XFEM and VCCT for delamination analysis of carbon fiber reinforced polymer (CFRP) laminate under mode-I loading for double cantilever beam (DCB), end-notch flexure (ENF), and mixed-mode bending (MMB) in 2020 [79]. Yin et al. used four-linear cohesive law for delamination simulation in multidirectional composite laminates in 2020 [80]. Garg et al. considered the scaled boundary finite element method (SBFEM) for delamination analysis of laminated composite using the cohesive zone method in 2020 [81]. Andraju and Raju simulated the progressive intra/inter-laminar damage in curved laminates under four-point bending load using cohesive zone model and progressive damage model in ABAQUS in 2020 [82]. Liang et al. implemented regularized extended finite element method (Rx-FEM) for progressive damage analysis of laminated composite in 2020 [83]. Gliszczynski and Wiącek performed an experimental and numerical analysis of end-notched flexure test of unidirectional carbon fiber reinforced composite laminates and modeled the fracture zone under mode-II loading conditions using virtual crack closure technique (VCCT) and cohesive zone method (CZM) in ANSYS in 2021 [84]. Raimondo et al. provided an experimental investigation to show the R-curve effect, and the effect of interface ply orientation on fracture toughness for double cantilever beam and mixed-mode bending specimen of unidirectional composite laminates and numerically examined R-curve behavior using VCCT in 2021 [85]. From previous and recent studies, it can be noticed that a comparison and combination of available finite element techniques and the effect of orientation on delamination is missing. Hence, the goal of this study is to overcome the gap in the study.

CHAPTER III

Theoretical Aspects

3.1 Finite Element Method

Practical design involves complex and non-linear characteristics. The finite element method can predict the deformation of solid shapes, displacement, and stress distribution under external load by solving partial differential equations. In solid mechanics, there are generally two types of finite element analysis. One is the analysis of static behavior of solid body where the sum of external force and internal force is zero and the motion of solid with time is not needed, which means $\sum F = 0$. Other is the determination of dynamic behavior in solids where the motion of a solid is calculated as a function of time, which means $\sum F = ma$. [86]

3.1.1 Finite Element Formulation for Linear Elasticity

Consider an original body R_0 and its boundary S_0, as shown in **figure 3.1**. A body force b is distributed, the displacement boundary condition $\mathbf{u}^*(\mathbf{x})$ is prescribed on a portion $\partial_1 R$ or traction on a portion $\partial_2 R$ of the boundary of R [86]. The applied boundary conditions on displacement and stress:

$$u_i = u_i^* \text{ on } \partial_1 R$$

$$\sigma_{ij} n_i = t_j^* \text{ on } \partial_2 R$$

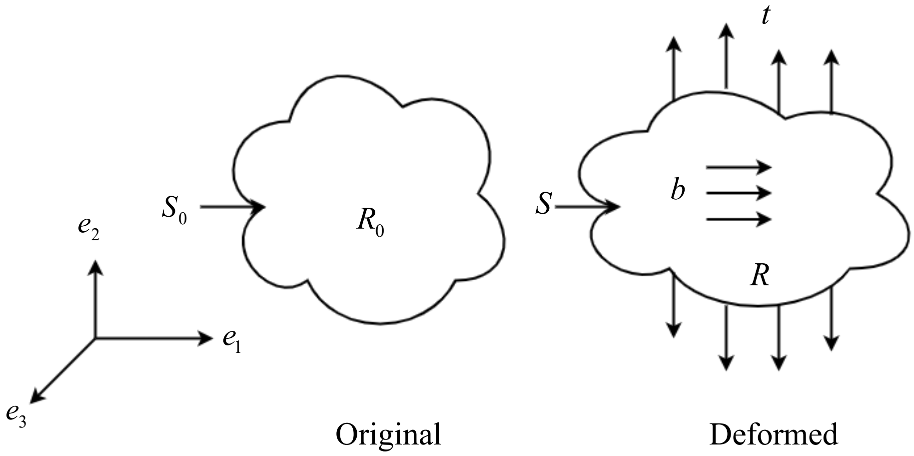


Figure 3.1: Geometric representation of boundary value problem.

The strain-displacement relationship can be expressed as:

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{3.1}$$

The elastic stress-strain law can be defined as:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} \tag{3.2}$$

The static equilibrium for stress can be presented as:

$$\frac{\partial \sigma_{ij}}{\partial x_i} + b_j = 0 \tag{3.3}$$

An associated virtual strain field can be defined as

$$\delta\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial \delta v_i}{\partial x_j} + \frac{\partial \delta v_j}{\partial x_i}\right) \tag{3.4}$$

According to the principle of virtual work, if the stress field σ_{ij} satisfies the equation given below for all possible virtual displacement fields and correlated virtual strain, that satisfies stress equilibrium $\frac{\partial \sigma_{ij}}{\partial x_i} + b_j = 0$ and the traction boundary condition $\sigma_{ij} n_i = t_j^*$ on $\partial_2 R$.

$$\int_R \sigma_{ij} \delta\varepsilon_{ij} dV_0 - \int_R b_i \delta v_i dV_0 - \int_{\partial_2 R} t_i \delta v_i dA = 0 \tag{3.5}$$

$$\int_R C_{ijkl} \frac{\partial u_k}{\partial x_l} \frac{\partial \delta v_i}{\partial x_j} dV - \int_R b_i \delta v_i dV - \int_{\partial_2 R} t_i^* \delta v_i dA = 0, \qquad u_i = u_i^* \text{ on } \partial_1 R \tag{3.6}$$

$\sigma_{ij} = \sigma_{ji}$ and that $\partial \delta\varepsilon_{ij} = \left(\frac{\partial \delta v_i}{\partial x_j} + \frac{\partial \delta v_j}{\partial x_i} \right) / 2$, so that

$$\sigma_{ij} \delta\varepsilon_{ij} = \left(\sigma_{ij} \frac{\partial \delta v_i}{\partial x_j} + \sigma_{ji} \frac{\partial \delta v_j}{\partial x_i} \right) / 2 = \sigma_{ij} \frac{\partial \delta v_i}{\partial x_j} \tag{3.7}$$

Finally, $\sigma_{ij} = C_{ijkl} \varepsilon_{kl} = C_{ijkl} \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) / 2$ and that the elastic compliances must satisfy $C_{ijkl} = C_{ijlk}$ so that $\sigma_{ij} = C_{ijkl} \frac{\partial u_k}{\partial x_l}$.

$$\sigma_{ij} \delta\varepsilon_{ij} = C_{ijkl} \frac{\partial u_i}{\partial u_j} \frac{\partial \delta v_i}{\partial x_j} \tag{3.8}$$

A finite element mesh is represented in **figure 3.2**. The displacement field is calculated at a set of n discrete points (called "nodes"). The coordinates of these unique points are denoted as x_i^a an unknown displacement vector at each nodal point will be u_i^a.

The displacement field at an arbitrary point can be denoted as:

$$u_i^b(x) = \sum_{a=1}^{n} N^a \left(x^b \right) u_i^a \tag{3.9}$$

The more traditional approach to choose interpolation functions, so that

$$N^a\left(x^b\right)=\begin{cases}1 & a=b\\ 0 & a\neq b\end{cases} \tag{3.10}$$

So, the virtual velocity can be interpolated as:

$$\delta v_i\left(x\right)=\sum_{a=1}^{n}N^a\left(x\right)\delta v_i^a \tag{3.11}$$

Substituting the interpolated fields into virtual work equation,

$$\int_R C_{ijkl}\frac{\partial N^b\left(x\right)}{\partial x_l}u_k^b\frac{\partial N^a\left(x\right)}{\partial x_j}\delta v_i^a dV-\int_R b_i N^a\left(x\right)\delta v_i^a dV-\int_{\partial_2 R} t_i^* N^a\left(x\right)\delta v_i^a dA=0 \tag{3.12}$$

In matrix form, the virtual work equation can be written as:

$$\left(K_{aibk}u_k^b-F_i^a\right)\delta v_i^a=0 \tag{3.13}$$

Where, global stiffness matrix and force matrix can be expressed as

$$K_{aibk}=\int_R C_{ijkl}\frac{\partial N^a\left(x\right)}{\partial x_j}\frac{\partial N^b\left(x\right)}{\partial x_l}dV \tag{3.14}$$

$$F_i^a=\int_R b_i N^a\left(x\right)dV+\int_{\partial_2 R} t_i^* N^a\left(x\right)dA \tag{3.15}$$

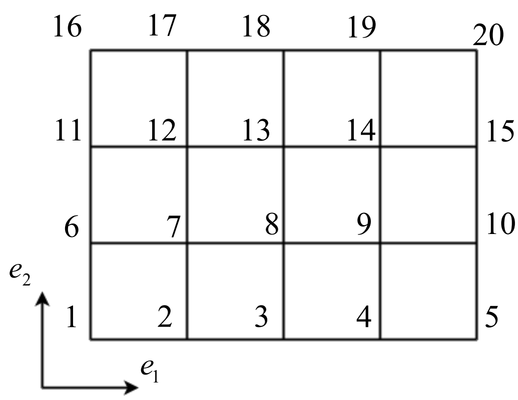


Figure 3.2: Representation of finite element mesh.

Interpolation function for each element in terms of a local and dimensionless coordinates system is defined as shape function, $N^a\left(\xi_i\right)$. The coordinate must need to satisfy $-1 \leq N^a\left(\xi_i\right) \leq +1$.

The displacement field and position of a point inside an element are computed in terms of the interpolation function as:

$$u_i = \sum_{a=1}^{N_e} N^a\left(\xi_i\right) u_i^a \tag{3.16}$$

$$x_i = \sum_{a=1}^{N_e} N^a\left(\xi_i\right) x_i^a \tag{3.17}$$

For an element, element stiffness matrix and force matrix,

$$k_{aibk}^{(l)} = \int_{V_e^{(l)}} C_{ijkl} \frac{\partial N^a\left(x\right)}{\partial x_j} \frac{\partial N^b\left(x\right)}{\partial x_l} dV \tag{3.18}$$

$$f_i^{a(l)} = \int_{V_e^{(l)}} b_i N^a\left(x\right) dV + \int_{\partial_2 V_e^{(l)}} t_i^* N^a\left(x\right) dA \tag{3.19}$$

After assemblage of each element's contribution, global stiffness and force can be modified as

$$K_{aibk} = \sum_{l=1}^{N_{lmn}} k_{aibk}^{(l)} \tag{3.20}$$

$$F_i^a = \sum_{l=1}^{N_{lmn}} f_i^{a(l)} \tag{3.21}$$

For unknown nodal displacement,

$$K_{aibk} u_k^b = F_i^a \quad \forall \left\{a,i\right\} : x_k^a \text{ not on } \partial_1 R \tag{3.22}$$

$$u_i^a = u_i^*\left(x_i^a\right) \quad \forall \left\{a,i\right\} : x_k^a \text{ on } \partial_1 R \tag{3.23}$$

2D interpolation function for four-noded quadrilateral element,

$$
\begin{aligned}
N^1 &= 0.25(1-\xi_1)(1-\xi_2) \\
N^2 &= 0.25(1+\xi_1)(1-\xi_2) \\
N^3 &= 0.25(1+\xi_1)(1+\xi_2) \\
N^4 &= 0.25(1-\xi_1)(1+\xi_2)
\end{aligned}
\tag{3.24}
$$

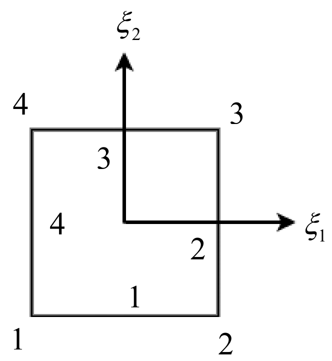


Figure 3.3: Four-noded quadrilateral element.

The shape function derivative can be written as

$$
\frac{\partial N^a}{\partial x_j} = \frac{\partial N^a}{\partial \xi_i}\frac{\partial \xi_i}{\partial x_j} \tag{3.25}
$$

So,

$$
\frac{\partial x_i}{\partial \xi_j} = \frac{\partial}{\partial \xi_j}\left(\sum_{a=1}^{N_e} N^a(\xi)\, x_i^a\right) = \sum_{a=1}^{N_e}\frac{\partial N^a(\xi)}{\partial \xi_j} x_i^a \tag{3.26}
$$

Inverse matrix of $\frac{\partial \xi_i}{\partial x_j}$ can be expressed as

$$
\frac{\partial \xi_j}{\partial x_i} = \left(\frac{\partial x}{\partial \xi}\right)_{ji}^{-1} \tag{3.27}
$$

Now,

$$J = \det\left(\frac{\partial x_i}{\partial \xi_i}\right) \tag{3.28}$$

Hence, global stiffness matrix and force matrix can be expressed as:

$$K_{aibk} = \int_{\Omega} C_{ijkl} \frac{\partial N^a(x)}{\partial x_j} \frac{\partial N^b(x)}{\partial x_l} J dV_{\xi} \tag{3.29}$$

$$F_i^a = \int_{\Omega} b_i N^a(x) J dV_{\xi} + \int_{\partial\Omega} t_i^* N^a(x) \hat{J} dA_{\hat{\xi}} \tag{3.30}$$

For the evaluation of this integrals, quadratic scheme is adopted as follows:

$$\int_{\Omega} f(\xi_i) dV_{\xi} = \sum_{I=1}^{N_I} w_I f(\xi_i^I) \tag{3.31}$$

So, the element stiffness matrix and force matrix can be denoted as:

$$k_{aibk} = \sum_{I=1}^{N_I} w_I C_{ijkl} \frac{\partial N^a(\xi_i^I)}{\partial \xi_p} \frac{\partial \xi_p}{\partial x_j} \frac{\partial N^b(\xi_i^I)}{\partial \xi_q} \frac{\partial \xi_q}{\partial x_l} J(\xi_i^I) \tag{3.32}$$

$$f_i^a = \sum_{I=1}^{N} w_I b_i N^a(\xi_i^I) J + \sum_{I=1}^{\hat{N}} w_I t_i^* N^a(\xi_i^I) \hat{J} \tag{3.33}$$

3.1.2 Finite Element Formulation for Incompatible Mode

Consider a double cantilever beam with length L, height $2a$, out-of-plane thickness, left end with resultant force P, and fixed right end, as shown in **figure 3.4** [86]. For thin beams, the four-noded quadrilateral elements cannot estimate the strain distribution due to bending accurately. This phenomenon, called shear locking, can be avoided by using a sufficiently fine mesh and an interpolation function that can estimate bending accurately. Incompatible mode elements contain accurate bending interpolation function that adds additional strain distribution to the element.

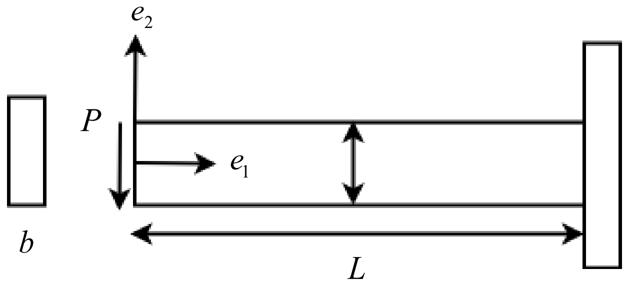


Figure 3.4: End-loaded cantilever beam.

The displacement fields in the element can be interpolated as

$$u_i = \sum_{a=1}^{N_e} N^a\left(\xi_i\right) u_i^a \tag{3.34}$$

$$\delta v_i = \sum_{a=1}^{N_e} N^a\left(\xi_i\right) \delta v_i^a \tag{3.35}$$

$$x_i = \sum_{a=1}^{N_e} N^a\left(\xi_i\right) x_i^a \tag{3.36}$$

The Jacobian matrix for the interpolation function can be written as:

$$\frac{\partial x_i}{\partial \xi_j} = \sum_{a=1}^{N_e} \frac{\partial N^a\left(\xi\right)}{\partial \xi_j} x_i^a \tag{3.37}$$

The determinant of the Jacobian matrix can be written as:

$$J = \det\left(\frac{\partial x_i}{\partial \xi_i}\right) \tag{3.38}$$

The inverse of the Jacobian matrix can be written as:

$$\frac{\partial \xi_j}{\partial x_i} = \left(\frac{\partial x}{\partial \xi}\right)_{ji}^{-1} \tag{3.39}$$

The displacement gradient in the element is expressed by

$$\frac{\partial u_i}{\partial x_j} = \left(\sum_{a=1}^{N_e} \frac{\partial N^a(\xi)}{\partial \xi_m} u_i^a + \sum_{k=1}^{p} \frac{J(0)}{J(\xi)} \alpha_i^{(k)} \delta_{km} \xi_k \right) \frac{\partial \xi_m}{\partial x_j} \tag{3.40}$$

Where, $p = 2$ for 2D and $p = 3$ for 3D.

Like displacement gradient, the virtual displacement gradient can be written as:

$$\frac{\partial \delta v_i}{\partial x_j} = \left(\sum_{a=1}^{N_e} \frac{\partial N^a(\xi)}{\partial \xi_m} \delta v_i^a + \sum_{k=1}^{p} \frac{J(0)}{J(\xi)} \alpha_i^{(k)} \delta_{km} \xi_k \right) \frac{\partial \xi_m}{\partial x_j} \tag{3.41}$$

3.1.3 Finite Element Formulation for Discontinuous Elasticity

Two adjacent surfaces cannot interpenetrate. Thus, $\Delta_n \geq 0$ [86]. The undamaged interface between two surfaces has stiffness, $k_0 = \frac{\sigma_{\max}}{d_1}$. When the separation between two surfaces or damage occurs, the damage internal variable, $D = 1$.

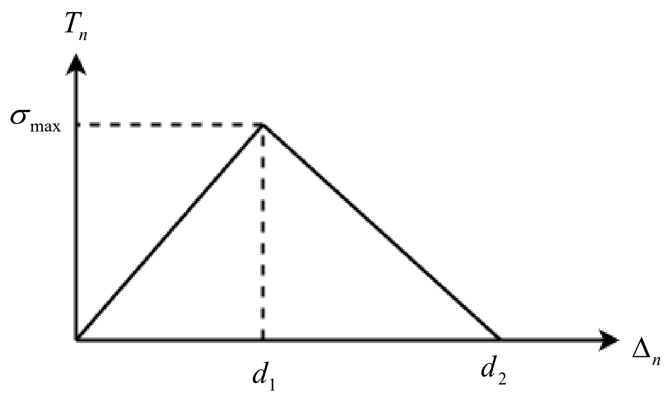


Figure 3.5: Traction-separation law.

Finite element governing equation for discontinuous surface in static case without considering body force can be written as:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \tag{3.42}$$

$$\sigma_{ij} = C_{ijkl} \frac{\partial u_k}{\partial x_l} \tag{3.43}$$

Boundary conditions for the body:

$\sigma_{ij} n_j = t_i^*$ on S_2 (Neumann boundary condition)

$u_i = u_i^*$ on S_1 (Dirichlet boundary condition)

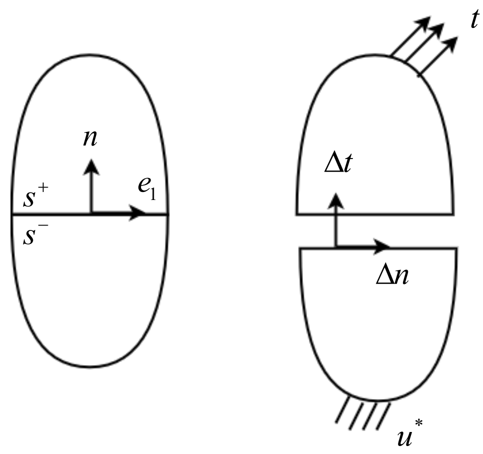


Figure 3.6: Interface separation surface.

Interface relations:

$$\begin{aligned} n_i \sigma_{ij} n_j &= T_n \text{ on } \Gamma_- \\ n_i \sigma_{ij} n_j &= -T_n \text{ on } \Gamma_+ \\ e_i^{\alpha} \sigma_{ij} n_j &= T_{\alpha} \text{ on } \Gamma_- \\ e_i^{\alpha} \sigma_{ij} n_j &= -T_{\alpha} \text{ on } \Gamma_+ \end{aligned} \tag{3.44}$$

Constitutive equation for cohesive zone can be defined as:

$$\left(T_n, T_{\alpha}\right) = f\left(\Delta_n, \Delta_{\alpha}\right) \tag{3.45}$$

Weak form of equilibrium can be denoted as:

$$\int_R \frac{\partial \sigma_{ij}}{\partial x_j} \eta_i dV = 0 \tag{3.46}$$

$$\text{Or, } \int_R \left\{ \frac{\partial}{\partial x_j}\left(\sigma_{ij}\eta_i\right) - \sigma_{ij}\frac{\partial \eta_i}{\partial x_j} \right\} dV = 0 \tag{3.47}$$

$$\text{Or, } \int_{\Gamma_-} \sigma_{ij} n_j \eta_i^- dA - \int_{\Gamma_+} \sigma_{ij} n_j \eta_i^+ dA + \int_S \sigma_{ij} n_j \eta_i dA - \int_R \sigma_{ij} \frac{d\eta_i}{dx_j} dV = 0 \text{; } \forall \text{ admissible } \eta_i \tag{3.48}$$

As $\Gamma \sigma_{ij} n_j = T_n \eta_i + T_\alpha e_i^\alpha$,

$$\int_R \sigma_{ij} \frac{d\eta_i}{dx_j} dV + \int_\Gamma T_n \left(\eta_i^+ - \eta_i^-\right) \eta_i dA + \int_\Gamma T_\alpha \left(\eta_i^+ - \eta_i^-\right) e_i^\alpha dA - \int_{S_2} t_i^* \eta_i dA = 0 \text{; } \forall \text{ admissible } \eta_i \tag{3.49}$$

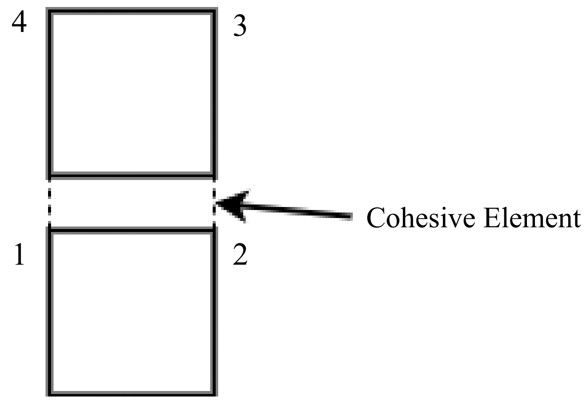


Figure 3.7: Representation of cohesive element.

Introducing interpolation function,

$$u_i = N^a u_i^a \tag{3.50}$$

$$\eta_i = N^a \eta_i^a \tag{3.51}$$

Discrete equations:

$$R_i^a \left[u_k^b \right] = f_i^a \tag{3.52}$$

Residual vector,

$$R_i^a = \sum_{el} \int_{\Omega el} \sigma_{ij} \left[u_k^b \right] \frac{\partial N^a}{\partial x_j} + \sum_{el} \int_{\Gamma el} T_n n_i \left[N^{a+} - N^{a-} \right] dA + \sum_{el} \int_{\Gamma el} T_\alpha e_i^\alpha \left[N^{a+} - N^{a-} \right] dA \tag{3.53}$$

Newton-Raphson iteration is required to solve the non-linear equation.

$$K_{aibk} dw_k^b = -R_i^a + f_i^a \tag{3.54}$$

$$u_k^b = u_{k-1}^b + dw_k^b \tag{3.55}$$

3.2 Techniques for Damage Analysis

ABAQUS is a commercial finite element software based on the finite element method. In ABAQUS, several modeling techniques like- the virtual crack closure technique, cohesive surface, cohesive element are used.

3.2.1 Virtual Crack Closure Technique

Delamination propagation has been assumed the same as crack propagation [87, 88] that delamination can be studied by the mathematics and science of fracture mechanics. Crack propagation is possible whenever the energy release rate per unit width and length of the damaged surface (called Strain Energy Release Rate) equals or is greater than the threshold limit or fracture toughness as:

$$G \geq G_c \tag{3.56}$$

Analytically, delamination has been studied by measuring strain energy release rate at the crack tip [89–91]. *Virtual Crack Closure Technique* (VCCT) is a finite element linear elastic fracture mechanics procedure that can be used to analyze the behavior of composite materials' structure in the presence of interlaminar damage, like delamination, in laminated composites and situations when force is propagated [92].

VCCT can be used only to analyze brittle crack propagation, and VCCT does not consider the energy dissipation due to plastic zone formation at the crack-tip [93]. Irwin first expressed the VCCT in 1957 [94], and the strain energy release rate is calculated using VCCT based on his prediction that the energy required to separate a surface equals the work required to close the surface during crack propagation. The total energy release rate is calculated from the nodal forces, and nodal displacements attained from the finite element mode, as illustrated in **figure 3.8** [95]. The virtual work done by force can be written as [96].

$$W = \frac{1}{2} F \left(\delta_u - \delta_l \right) \tag{3.57}$$

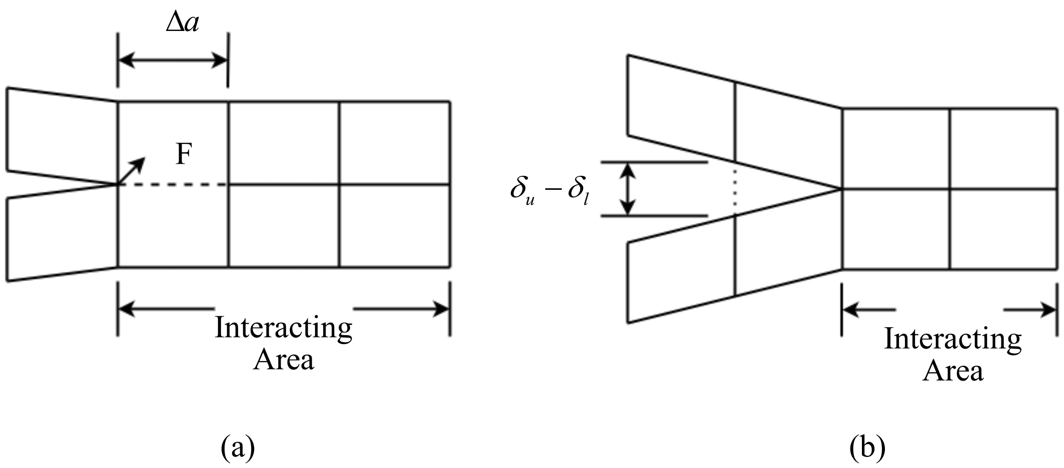


Figure 3.8: Crack closure technique (a) Crack is closed; (b) Crack is extended.

The total energy release rate can be written as [96].

$$G = \lim_{\Delta a \to 0} \frac{W}{\Delta A} = \lim_{\Delta a \to 0} \frac{1}{2} F \frac{\delta_u - \delta_l}{\Delta A} \tag{3.58}$$

To begin with, the work done W needed to close the crack can be determined in two analyses. First, the stress at the crack tip must be evaluated for a crack length a, and second, the displacement of the configuration is determined with a crack tip advanced from a to $a + \Delta a$. By assuming crack extension has a negligible effect on the crack tip, stress and displacement can be estimated in only one analysis. From this concept, the work W required to close the crack can be expressed as [92].

$$W = \frac{1}{2}\left[\int_0^{\Delta a} \sigma_{yy}^{(a)}(x)\,\delta u_y^{(a)}(x-\Delta a)\,dx + \int_0^{\Delta a} \sigma_{yx}^{(a)}(x)\,\delta u_x^{(a)}(x-\Delta a)\,dx + \int_0^{\Delta a} \sigma_{yz}^{(a)}(x)\,\delta u_z^{(a)}(x-\Delta a)\,dx\right] \quad (3.59)$$

Combining this expression of equation **3.59** with the energy release rate of equation **3.58**, the strain energy release rate for three orthogonal fracture modes: G_{I}, G_{II}, and G_{III}, can be obtained for opening mode I, in-plane shear mode II and anti-plane shear (tear) mode III, respectively.

The strain energy release rate, the work done by the nodal forces to close the crack tip in the finite element model, as shown in **figure 3.9**, for three orthogonal modes, is expressed as:

$$\begin{aligned} G_{\mathrm{I}} &= \lim_{\Delta a \to 0} \frac{1}{2b\Delta a} F_y^{cd}\left(u_2^c - u_2^d\right) \\ G_{\mathrm{II}} &= \lim_{\Delta a \to 0} \frac{1}{2b\Delta a} F_x^{cd}\left(u_1^c - u_1^d\right) \\ G_{\mathrm{III}} &= \lim_{\Delta a \to 0} \frac{1}{2b\Delta a} F_{yz}^{cd}\left(u_3^c - u_3^d\right) \end{aligned} \quad (3.60)$$

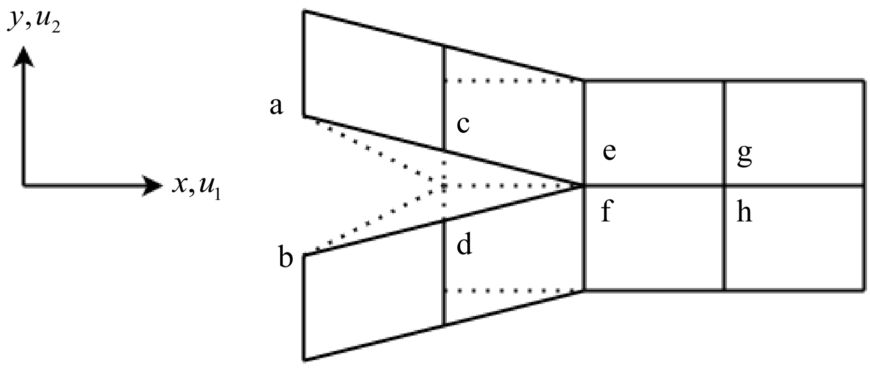


Figure 3.9: Calculation of energy release using Virtual Crack Closure technique.

Thus, the total energy release rate can be expressed as:

$$G = G_{\mathrm{I}} + G_{\mathrm{II}} + G_{\mathrm{III}} \quad (3.61)$$

3.2.2 Cohesive Zone Method

The *Cohesive Zone Method* (CZM) is another finite element approach for numerical simulation of delamination based on cohesive damage mechanics (dependent on the cohesive damage zone near the crack tip) [92]. Dugdale and Barenblatt have proposed CZM, and they disclosed that the value of stresses is zero at the crack tip, and stresses are progressively developed at the partially damaged region of material called the cohesive zone [97, 98]. Hillerborg et al. first used the cohesive zone approach to analyze crack initiation and propagation under mode-I loading conditions [99]. This approach assumes that the force transfer capability between two split laminates of delamination does not lose completely through damage initiation, but the stiffness of the interface between two laminates reduces progressively [93]. Cohesive zone modeling is applied to the finite element model by assigning elements along the damage path [100].

3.2.2.1 Cohesive Element

The cohesive element method is based on elements with a finite thickness used to analyze the behavior of adhesive bonds during delamination with considerable interface thickness. This approach is based on classical engineering stress-strain (σ-ε) behavior. [93]

3.2.2.2 Cohesive Surface

The cohesive surface method is based on elements with zero thickness used to analyze the behavior of adhesive bonds during delamination with negligible interface thickness. This approach is based on engineering traction-separation (σ-Δ) behavior. The cohesive element with zero thickness consists of coincident opposite nodes of adjacent laminates, and the adjacent laminates separate during delamination of the laminated composites as the opposite coincident nodes are separate entities. The separation of adjacent laminates is proportional to the stiffness of the cohesive element as adjacent laminates are connected due to the stiffness of the composite laminates. [93]

In the most straightforward and practical concept of CZM, the entire body is considered linear-elastic, and the interface or the cohesive zone is considered with non-linear cohesive law [92]. The cohesive law links the traction with displacement along the crack plane. The stiffness of the cohesive element is decreased, and the traction in the cohesive zone is reduced to zero when the area under the traction-displacement curve is equal to the fracture toughness [101]. The complete fracture process consists of four stages, as shown in **figure 3.10**. In stage 1, the model behaves like linear elastic same as the assumption. In stage 2, crack initiates as the load increases at the crack tip. Stage 2 and 3 cover an area controlled under the cohesive law that is non-linear and progresses from crack initiation to propagation. In stage 4, the area under the traction-separation curve equals the fracture toughness and new traction-free surface forms. For finite element modeling using cohesive zone modeling, finite bulk elements are required for modeling stage 1 generating boundary with cohesive surface elements for modeling crack initiation (stage-2), crack evolution (stage-3), and complete failure (stage-5).

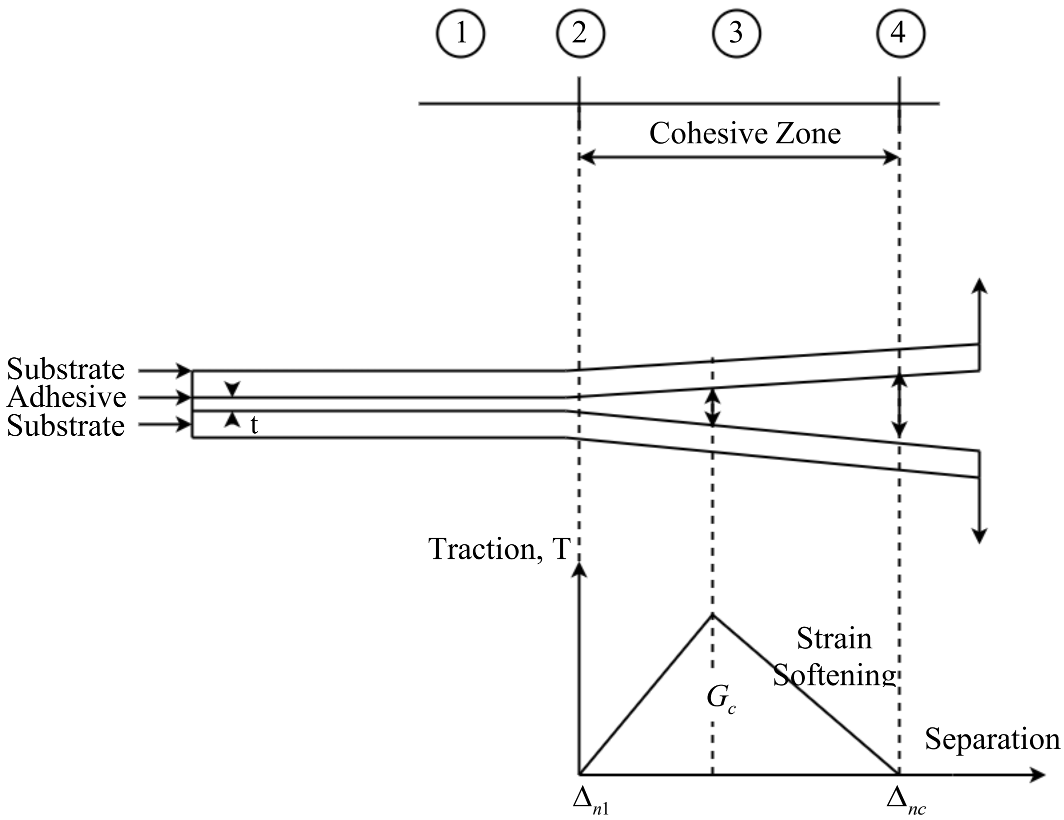


Figure 3.10: Cohesive zone modeling and bi-linear traction-separation law in an adhesively bonded joint.

According to the cohesive law, cohesive fracture energy can be expressed as:

$$G_c = \frac{1}{2}\sigma_c \Delta_{nc}\left(1-\lambda_1\right) \tag{3.62}$$

According to the cohesive law, displacements when crack initiates and evolutes can be expressed as:

$$\Delta_{n1} = \lambda_1 \Delta_{nc} \tag{3.63}$$

Quadratic nominal stress criterion (Quads), a damage initiation criterion for traction separation in ABAQUS, can be expressed as:

$$f = \left\{\frac{\langle t_n \rangle}{t_n^0}\right\}^2 + \left\{\frac{t_s}{t_s^0}\right\}^2 + \left\{\frac{t_t}{t_t^0}\right\}^2 \tag{3.64}$$

It is assumed that the damage is initiated when $f = 1$.

Benzeggagh and Kenane, a damage initiation criterion for traction separation in ABAQUS, can be expressed as:

$$\frac{G_T}{G_{\mathrm{I}c} + \left(G_{\mathrm{II}c} - G_{\mathrm{I}c}\right)\left(\dfrac{G_{\mathrm{II}} + G_{\mathrm{III}}}{G_T}\right)^{\eta}} \geq 1 \tag{3.65}$$

CHAPTER IV

Modeling Method and Steps

4.1 Modelling Steps

Finite element modeling consists of several fundamental steps, as shown in **figure 4.1**. The first step involves geometry and modeling, where a 2D model with appropriate dimensions is created. Then material properties are assigned to the geometry. After that, the discrete parts are assembled into a single geometry. Then the model is discretized to generate finite element mesh. Boundary conditions are applied, and interaction between parts is defined. Geometrical nonlinear static general solver is used for finite element analysis. Then mesh sensitivity analysis is performed to find out the appropriate mesh. After passing the mesh sensitivity, the result is extracted from the finite element software.

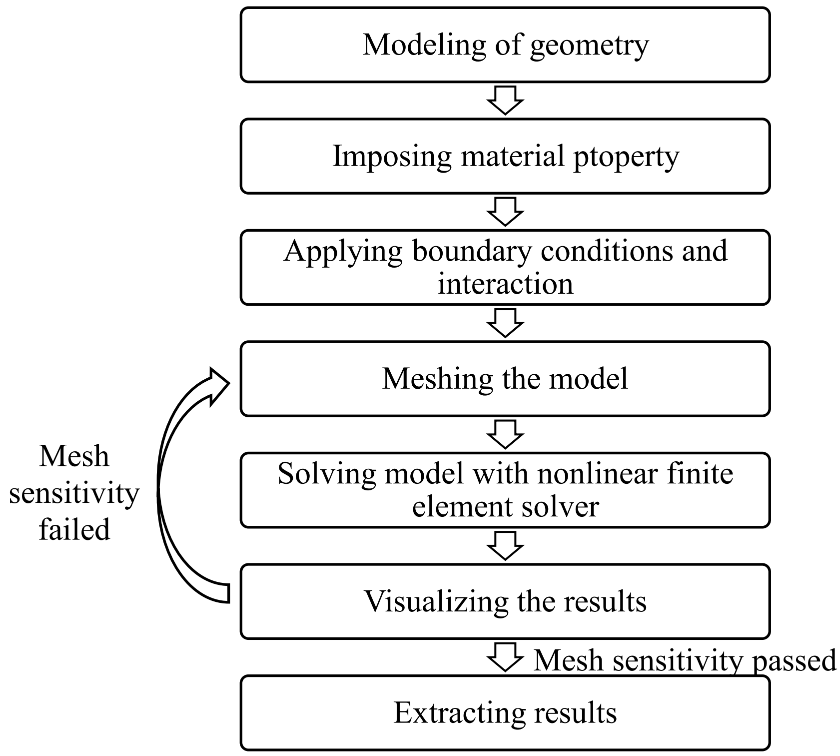


Figure 4.1: Fundamental steps of modeling and simulation in finite element method.

4.2 Geometry of the Model

For finite element analysis of delamination in ABAQUS, a double cantilever beam (DCB) with 100 mm length and 1.5 mm thickness of each is modeled with 30 mm initial crack length, as presented in **figure 4.2**.

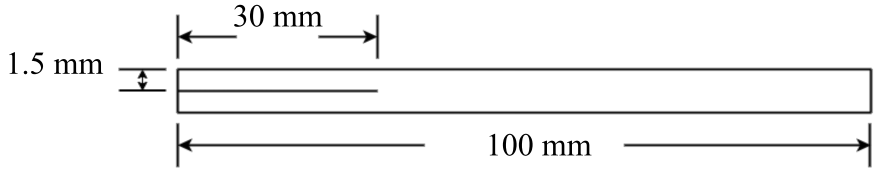


Figure 4.2: Geometry of double cantilever beam (DCB) with dimension.

4.3 Material Properties

The modeled double cantilever beam consists of two sections: One is composite, and another is adhesive. Carbon fiber-reinforced composite (CFRP) properties are assigned as the material definition for the composite bulk section, as represented in **table 4.1** [93].

Table 4.1: Material properties of composite laminates.

Longitudinal elastic modulus, E_1 (GPa)	135.3
Transverse in-plane elastic modulus, E_2 (GPa)	9.0
Transverse out-of-plane elastic modulus, E_3 (GPa)	9.0
Major in-plane Poisson's ratio, ν_{12}	0.24
Out-of-plane Poisson's ratio, ν_{13}	0.24
Out-of-plane Poisson's ratio, ν_{23}	0.46
In-plane shear modulus, G_{12} (GPa)	4.5
Out-of-plane shear modulus, G_{13} (GPa)	3.3
Out-of-plane shear modulus, G_{23} (GPa)	4.5

Table 4.2: Material properties for adhesive

E / K_{nn} (Pa)	5.7E14
$G1 / K_{tt}$ (Pa)	5.7E14
$G2 / K_{ss}$ (Pa)	5.7E14
σ_{I}^{0}	5.7E7
σ_{II}^{0}	5.7E7
σ_{III}^{0}	5.7E7
G_{Ic} (J/m^2)	280
G_{IIc} (J/m^2)	280
G_{IIIc} (J/m^2)	280

The adhesive section is modeled for three approaches (Cohesive element, Cohesive surface, and VCCT) with similar material properties for each method, presented in **table 4.2** [93].

4.4 Boundary Conditions

The displacement boundary condition of 6 mm is applied to the upper left end, and the lower left and other ends of the model are fixed, as shown in **figure 4.3**.

Figure 4.3: Applied boundary condition on the double cantilever beam.

4.5 Finite Element Model

This study modeled DCB for delamination analysis using VCCT as shown in **figure 4.4**, cohesive surface as shown in **figure 4.4**, and cohesive element as shown in **figure 4.5**.

Figure 4.4: Finite element modeling of DCB for delamination analysis using VCCT and cohesive surface.

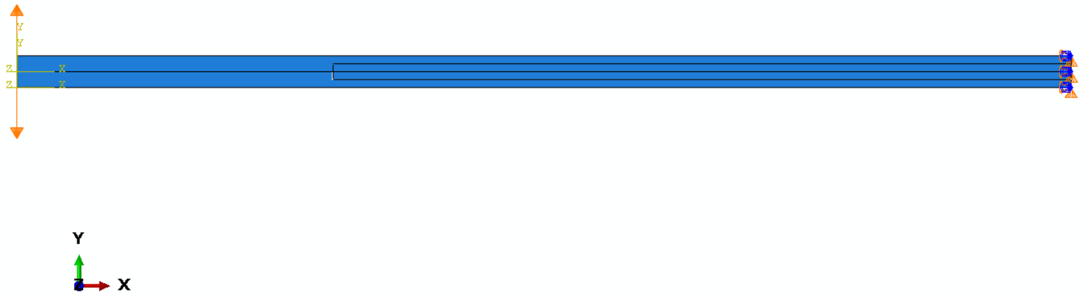


Figure 4.5: Finite element modeling of DCB for delamination analysis using cohesive element.

4.6 Model mesh sensitivity analysis

The finite element 2D composite material model contains four-node quadrilateral elements (CPE4I). For model #1, a coarser mesh with 135 elements is used. For consecutive model #2; model #3; model #4 and model #5, 270; 540; 1080 and 2160 elements are used, as shown in figure 4.6. In model #5, a finer mesh is considered than model #4; however, both models' reaction force is close to each other. Model #4 is computationally economical for simulation, whereas model #5 is computationally expensive, and using coarser mesh than model #4 challenges the computational accuracy. Hence, model #4 with 1080 elements is employed in this study, as shown in figure 4.7.

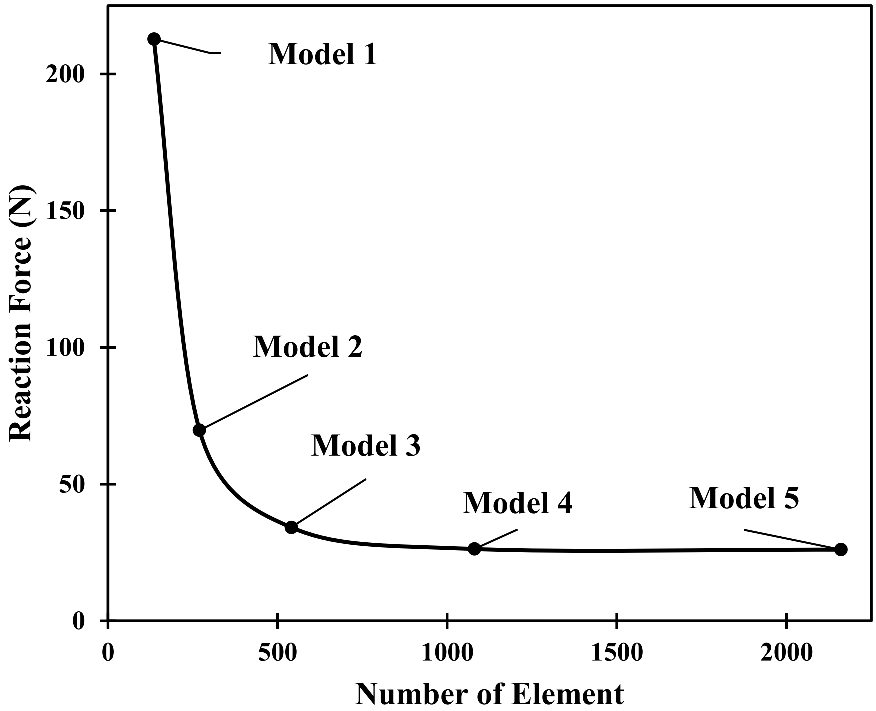


Figure 4.6: Mesh Sensitivity Analysis.

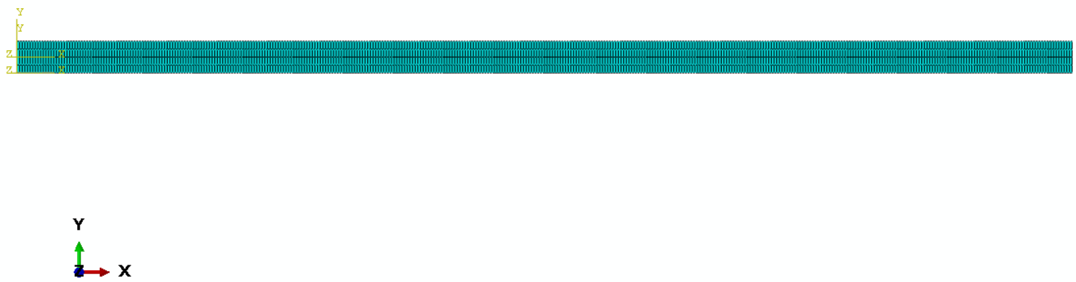


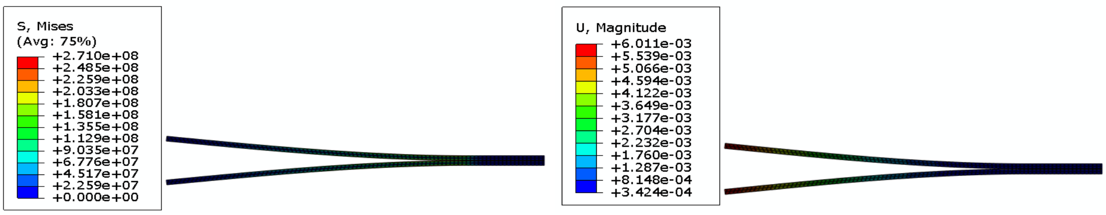
Figure 4.7: Employed mesh for modeling.

CHAPTER V

Result and Discussion

5.1 Effect of Mode-I Loading on Delamination Initiation and Propagation

In the cohesive element approach, a double cantilever beam has been modeled with dimension as **figure 4.2** with cohesive elements of finite thickness of 1.5 mm with 70 mm length and 20 mm width. Non-linear geometry has been implemented, and the cohesive elements have been tied with the beam. The material properties for the beam are identical to **table 4.1**, and the material properties for adhesive are similar to **table 4.2**. The displacement boundary condition of 6 mm has been implemented to the lower left end, and the upper left end and load displacement curve has been obtained, as shown in **figure 5.2**.



(a) (b)

Figure 5.1: The (a) stress-field and (b) displacement for delamination in the cohesive element model.

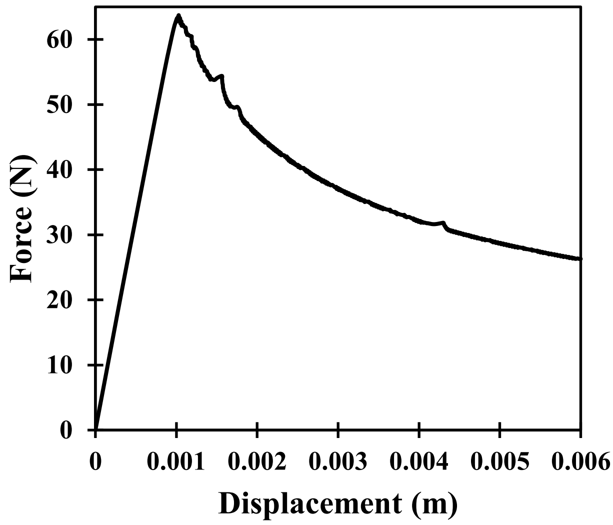
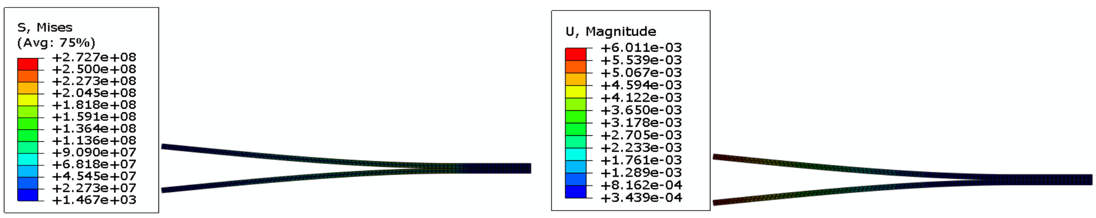


Figure 5.2: Force vs. displacement curves for cohesive element approach.

In the cohesive surface approach, a double cantilever beam with a similar dimension as **figure 4.2** has been modeled with cohesive elements of zero thickness, which means surface to surface contact is established at the interface of the beam. Non-linear geometry has been applied. There is a master surface and a slave surface, and material properties as **table 4.2** are assigned at the contact. As a result of using the 6 mm displacement boundary condition, a load-displacement plot has been attained, as shown in **figure 5.4**.



(a) (b)

Figure 5.3: The (a) stress-field and (b) displacement for delamination in the cohesive surface model.

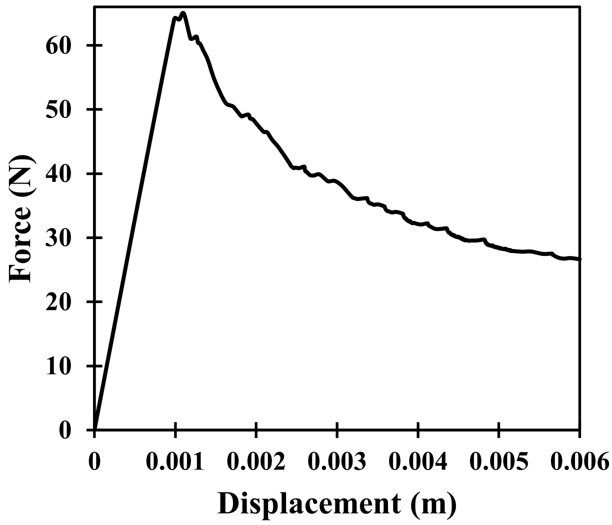
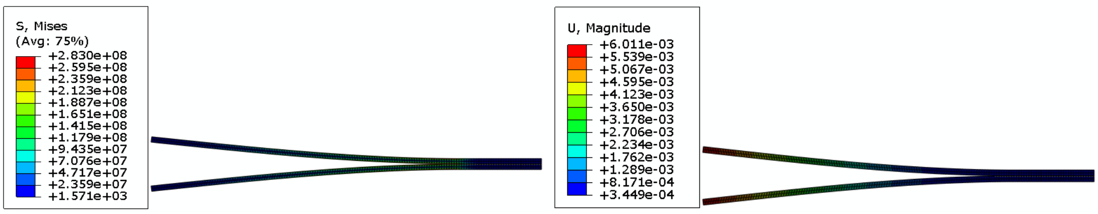


Figure 5.4: Force vs. displacement curves for cohesive surface method.

In the VCCT approach, a similar beam model with the same dimension has been modeled, and non-linear geometric analysis has been used. Similar property has been applied to the adhesive that is the contact surfaces of the beam. A similar boundary condition has been applied, and the load-displacement curve is obtained, as shown in **figure 5.6**.



(a) (b)

Figure 5.5: The (a) stress-field and (b) displacement for delamination in the VCCT model.

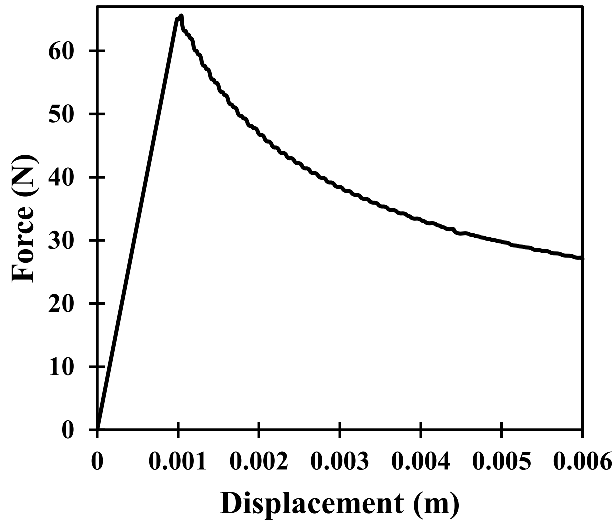


Figure 5.6: Force vs. displacement curves for VCCT.

The load-displacement curve of a double cantilever beam for three different approaches (VCCT, cohesive element, cohesive surface) is compared, as shown in **figure 5.7**. These three approaches can model delamination efficiently but cannot model non-linear fiber bridging. Similar results are obtained from comparing the load-displacement curve for delamination propagation for three methods.

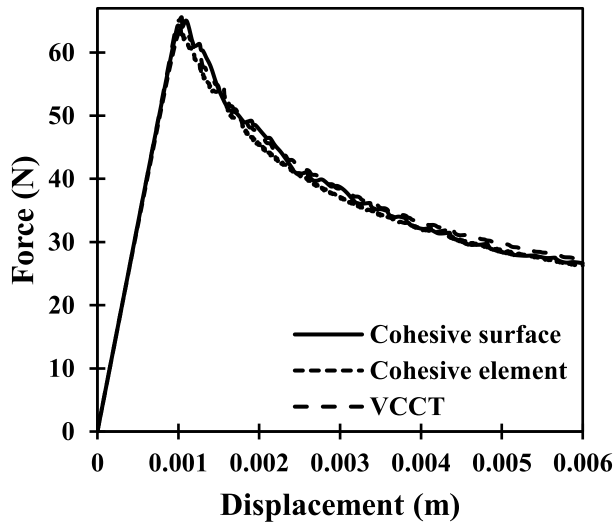


Figure 5.7: Comparison of force vs. displacement curves for three different finite element modeling of delamination

5.2 Validation of Numerical Solution with Analytical Solution

5.2.1 Numerical Solution

Due to the application of mode-I loading, the delamination has been initiated and propagated for each model, as presented in **figure 5.8**.

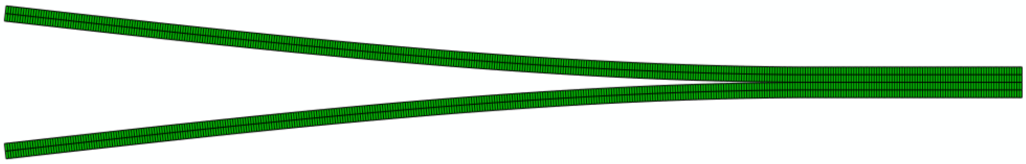


Figure 5.8: Numerical visualization of delamination of composite.

From **figure 5.7**,

The critical load when the delamination is initiated in the VCCT model = 65.6 N,

The critical load in the cohesive element model = 63.7 N, and

The critical load for the cohesive surface model = 64.9 N.

5.2.2 Simplified Analytical Solution

Consider the top arm of a double cantilever with the initial crack length a_0 for the analytical solution.

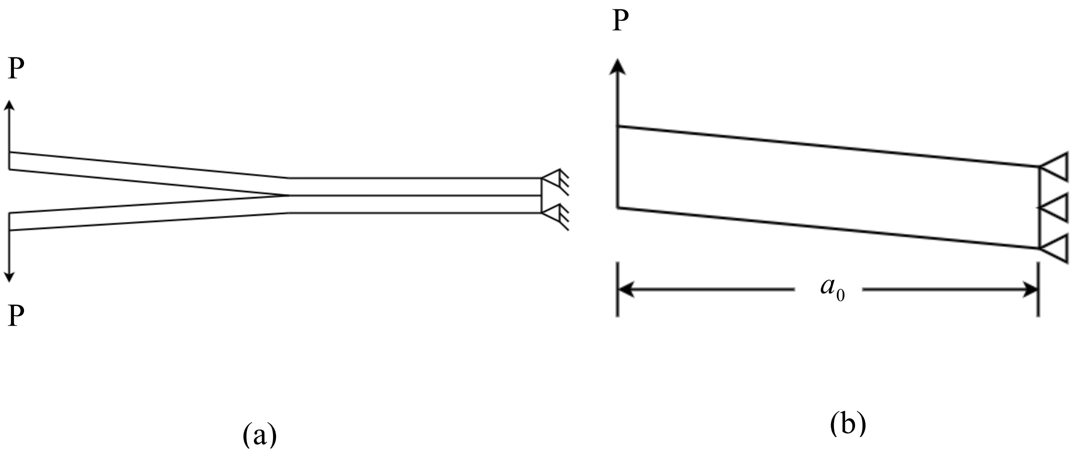


Figure 5.9: (a) Double cantilever beam (b) Top arm of the beam.

The deflection of a cantilever beam can be expressed as

$$\delta = \frac{Pa_0^{\ 3}}{3EI} = \frac{Pa_0^{\ 3}}{3E\dfrac{Bh^3}{12}} = \frac{4Pa_0^{\ 3}}{EBh^3}$$

The strain energy before crack, $U_B = 0$, as equal and opposite two forces are applied to the left end of the double cantilever beam, as shown in **figure 5.9(a)**.

The strain energy after crack, $U_A = \dfrac{4P^2a_0^{\ 3}}{EBh^3}$

The energy release rate for mode-I can be expressed as

$$G_I = \frac{dU}{dA} = \frac{1}{B}\frac{dU}{da}$$

$$\text{Or, } G_I = \frac{1}{B}\frac{d}{da}\left(\frac{4P^2a_0^{\ 3}}{EBh^3} - 0\right)$$

$$\text{So, } G_I = \frac{12P^2a_0^{\ 2}}{EB^2h^3}$$

When the energy release rate is equal to the critical energy release rate, force will be equal to the critical load. That means, $G_I = G_{Ic}$, $P = P_c$.

For modeled double cantilever beam,

Width of the beam, B = 20 mm = 0.02 m

Thickness of the beam, h = 1.5 mm = 0.0015 m

Initial crack length, a_0 = 30 mm = 0.03 m

Young's modulus, E = 120 E 9 Pa (considering isotropic and taking average)

Fracture toughness or critical energy release rate, G_{Ic} = 280 J/m^2

The critical load or maximum load that can be withstand by the beam when crack initiated can be expressed as

$$P_c = \left(\frac{G_{Ic}EB^2h^3}{12a_0^{\ 2}}\right)^{\frac{1}{2}}$$

Hence, analytical critical load for the model when delamination starts is

$$P_c = \left(\frac{280 \times 120 \times 10^9 \times (0.02)^2 \times (0.0015)^3}{12 \times (0.03)^2} \right)^{\frac{1}{2}} = 64.8 \text{ N}$$

The analytically calculated critical load is close to the numerically obtained loads.

5.3 Effect of Energy on Modelling Techniques

According to the law of conservation of energy, external energy is equal to the sum of internal energy and energy dissipation for delamination. The applied displacement boundary condition is the external energy, and the internal energy leads to the strain. The energy difference between external and internal energy is responsible for delamination in VCCT, and the relation between external energy, internal energy, and damage energy and displacement is represented in **figure 5.10**. It is impossible to get back the dissipated energy, but one way to get back that energy is by using a self-healing agent at the delamination interface.

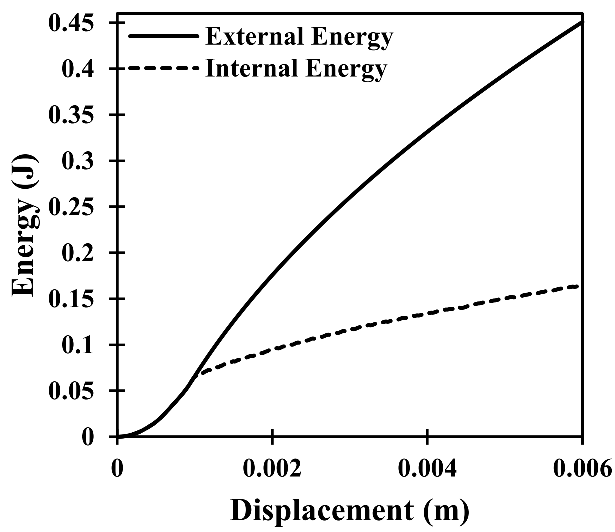


Figure 5.10: Energy vs. displacement curve for delamination propagation in VCCT.

In the case of cohesive surface and cohesive element, damage occurs following traction-separation law that no energy dissipation occurs due to delamination. Hence, external energy and internal energy are identical, as shown in **figure 5.11** and **figure 5.12** respectively.

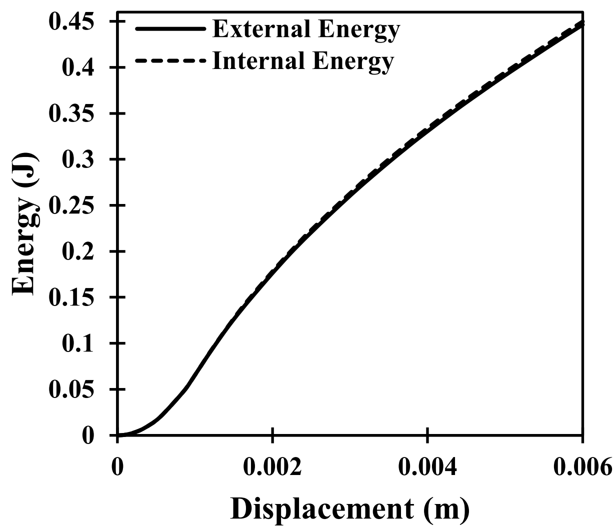


Figure 5.11: Energy vs. displacement curve for delamination propagation in cohesive surface.

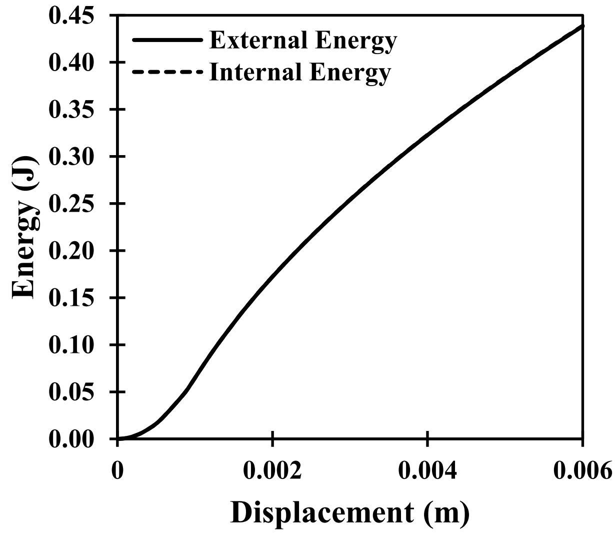


Figure 5.12: Energy vs. displacement curve for delamination propagation in cohesive element.

5.4 Effect of Crack Opening Displacement on Crack Length

The crack length increases with the crack opening displacement at the crack tip, as shown in **figure 5.13** and similar crack length for increasing crack opening displacement at the crack tip is obtained for three different approaches.

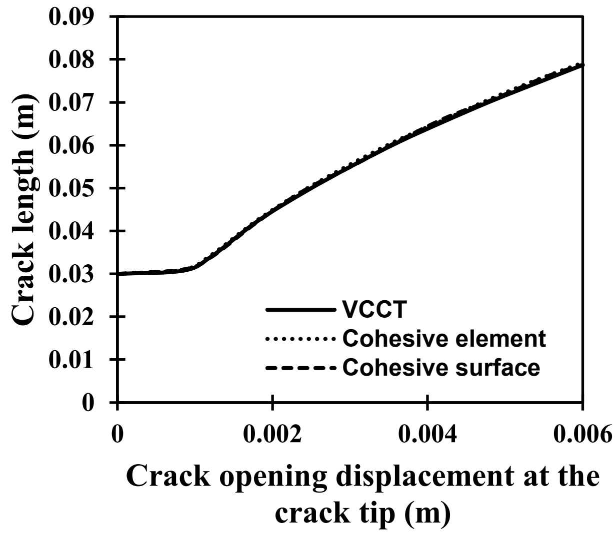


Figure 5.13: Crack length vs. displacement for delamination propagation.

5.5 Computational Time Analysis

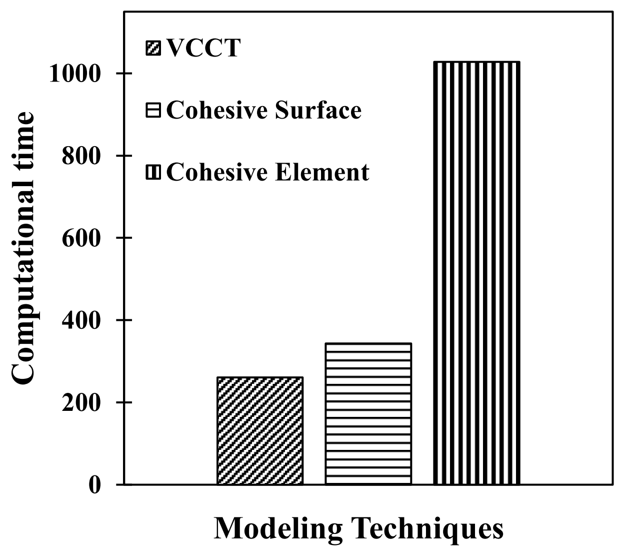


Figure 5.14: Representation of computational time requirement.

The computational time for cohesive element, cohesive surface, and VCCT is 1028 seconds, 343 seconds, and 261 seconds respectively (all the simulations have been done on a personal computer with Core i5 CPU and 8 GB RAM). The least time is required for the VCCT technique, as shown in **figure 5.14**.

5.6 Effect of Material Orientation on Delamination Propagation

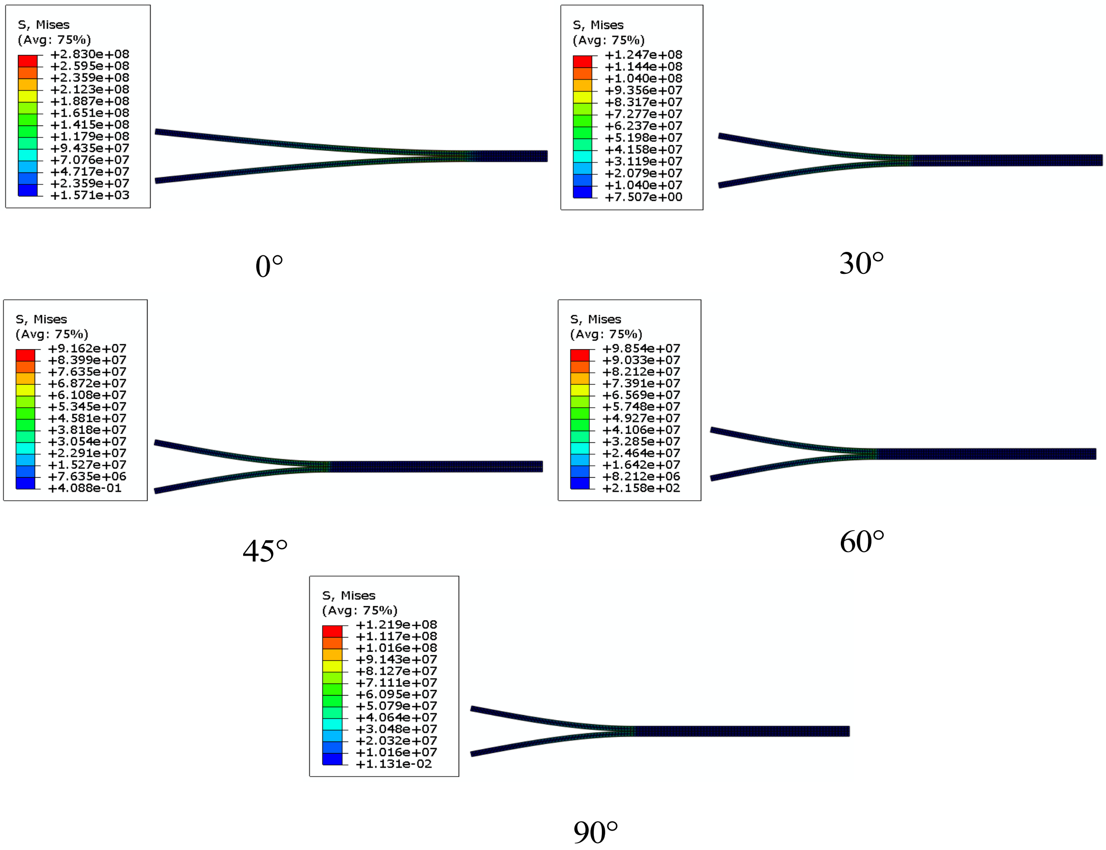


Figure 5.15: Variation of stress field with different orientation.

The delamination opening decreases from 0° to 90° orientation because the elastic properties of the composite change from longitudinal to the transverse direction, which prevents delamination and improves interlaminar strength, as shown in **figure 5.15** and **figure 5.16**. However, changing elastic properties from longitudinal to transverse direction is detrimental to the overall tensile property of composite.

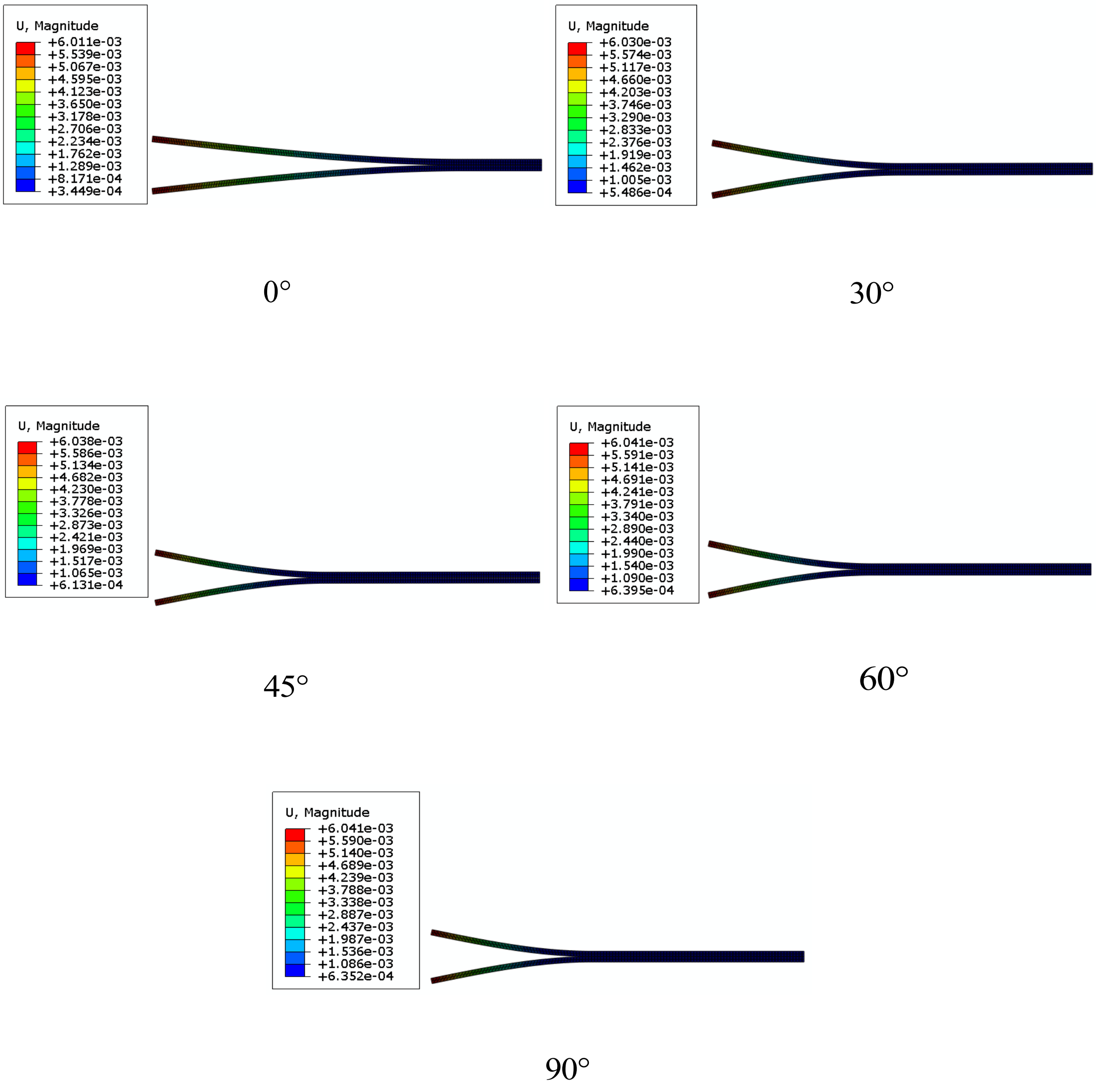


Figure 5.16: Variation of displacement with different orientation.

At 0° orientation, delamination initiates for the minor crack opening displacement, whereas delamination starts with higher crack opening displacement at 90° orientation, as shown in **figure 5.17**. Hence, an optimum orientation needs to use for preventing delamination and better tensile property together.

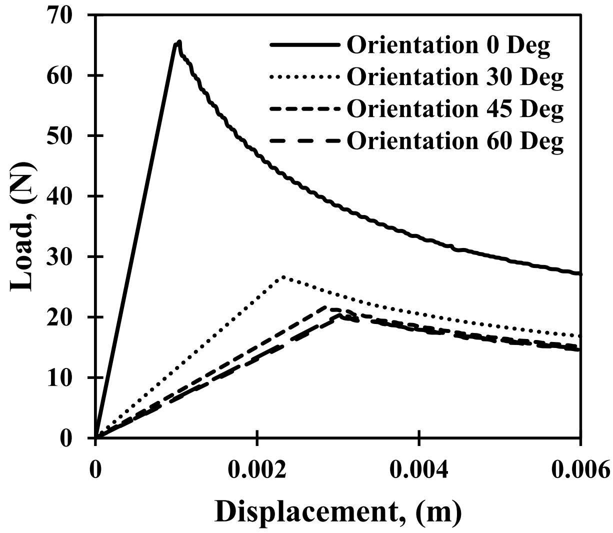


Figure 5.17: Variation of load with different orientations.

CHAPTER VI

Conclusion and Future Work

6.1 Conclusion

Delamination is the most occurred failure of composite in service; that delamination has been analyzed numerically using VCCT, cohesive surface technique, and cohesive element approach. A double cantilever beam is modeled for delamination analysis, and the findings from the analysis are as follows:

- Delamination initiation and propagation start at the same displacement and force in these three techniques.
- The critical load obtained analytically and numerically is almost the same.
- VCCT requires the least computational time, though VCCT requires the complex calculation of fracture parameters from nodes behind and ahead from the crack tip. However, VCCT provides similar results as the cohesive element approach and cohesive surface. By analyzing with the cohesive element method, it can be noticed that the longest computational time is required as the geometry of the cohesive element model used for analysis is more complex than others due to the presence of the layer of cohesive element.
- The external energy from the applied displacement boundary condition is divided into two parts: one part of energy remains internal energy causing strain, and another part of energy dissipates for crack propagation. The dissipated energy can be recovered using a healing agent containing capsule, particulate, or fiber at the interface along which the crack propagates. In cohesive surface and cohesive element, the dissipated energy is calculated as a sum with internal energy following the formula, so external energy and internal energy are the same. In the case of VCCT, the energy dissipation is calculated separately following the formula that internal energy is less than internal energy.
- The crack length increases with crack opening displacement and similar crack length for increasing crack opening displacement at the crack tip is obtained for three different approaches.

- For 0° orientation, delamination initiates under minimum crack opening displacement, and maximum crack opening displacement is required at 90° orientation. However, a change of elastic behavior at the 90° orientation is not a good choice where the tensile property is required.

6.2 Future work

Recently, the self-healing approach has become a demanding sector of research as self-healing is a determined way to ameliorate the damage bearing capability and extend the life-span of constructional materials. The materials that can cure decay within themselves or under outward incentive are called self-healing materials [102], and Self-healing for cracks expands the durability of the material. The self-healing technique is trendy for the researcher to mitigate the crack propagation in the materials. In the future, the analysis of damage and healing of materials with self-healing phenomena using the extended finite element method (XFEM) with cohesive zone model (CZM) will be performed.

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